

SHUBHAM GAJJAR

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EDUCATION

Northeastern University, Portland, Maine

September 2025 – May 2027

Master of Science in Artificial Intelligence, Grade Point Average: 4.0/4.0

LDRP Institute of Technology and Research, Gandhinagar, India

September 2022 – May 2025

Bachelor of Engineering in Computer Engineering, Grade Point Average: 8.41/10.0

Relevant Coursework: Foundations of Artificial Intelligence, Machine Learning, Research Capstone, Applied Programming and Data Processing for AI, Deep Learning, Computer Vision, Algorithms, Image Processing

TECHNICAL SKILLS & CERTIFICATION

Programming Languages: Python, JavaScript

Deep Learning: PyTorch, TensorFlow, Keras, HuggingFace Transformers, CUDA (GPU acceleration), QLoRA

Computer Vision & Multimodal: OpenCV, Albumentations, Matplotlib, MedGemma, MedSigLIP, DINOv3, CLIP

Data Science & MLOps: NumPy, Pandas, Scikit-learn, MLflow, PySpark, Docker, PostgreSQL

Web Development & Tools: Next.js, React Native, FastAPI, Tailwind CSS, Git

Certificates: Python for Data Science from Indian Institute of Technology Madras, Python Data Structures from University of Michigan

PROFESSIONAL EXPERIENCE

Northeastern University, Portland, Maine

Research Assistant

February 2026 – Present

- Designed end-to-end vision-language model fine-tuning pipeline using Google MedGemma 4B (SigLIP vision encoder, Gemma-3 backbone) with QLoRA (4-bit NF4 quantization, LoRA adapters on attention projections, trainable language-model head, frozen vision encoder) for cell tumor classification from microscopy
- Built preprocessing pipeline merging multi-channel microscopy into normalized RGB images resized to 512×512, generating a 5-prompt visual question answering dataset from expert-reviewed annotations on a 70:20:10 split with in-memory caching via batched PySpark reads
- Trained on NVIDIA H100 GPU with label masking (loss on response tokens only), tracking token accuracy, cross-entropy loss, ROUGE-L, and BERTScore through MLflow with versioned model artifacts

BigCircle (UPSAAS Technologies LLP), Gandhinagar, India

Artificial Intelligence Engineer

January 2025 – August 2025

- Developed end-to-end Deep Research pipeline in Python orchestrating ML prompt generation, Firecrawl API for web scraping, and ChatGPT API summarization with graph visualization via Matplotlib
- Automated Typst PDF report generation and optimized API call efficiency, reducing processing time by 75%
- Engineered pagination and authentication systems using JavaScript and Next.js, accelerating page load times by 40%

KEY PROJECTS & PUBLICATIONS

Next-Step: Full-Stack School Management Platform

Academic Project

May 2026

- Built async REST API with FastAPI exposing 20+ endpoints across 10 router modules backed by 9 SQLAlchemy ORM models (cascade deletes, JSON columns, unique constraints) with Alembic migrations on PostgreSQL and Celery/Redis with Flower monitoring for background tasks
- Implemented JWT authentication using Argon2 password hashing and Google OAuth 2.0 with PKCE flow, supporting Admin, Teacher, and Student roles
- Crafted cross-platform mobile app (iOS, Android, Web) with React Native and Expo using TypeScript, Zustand state management, file-based routing, and Google Drive integration for document management

MorphoCLIP: Cell Microscopy and Text Contrastive Learning

Research Capstone

April 2026

- Led 3-person team building dual-encoder contrastive learning system aligning cell microscopy with natural-language perturbations, achieving 24.3% Recall@10 on 817-perturbation retrieval, 3.3× over CellCLIP using 8M parameters versus 1.48B (185× reduction)
- Architected DINOv3 vision backbone paired with BioClinical ModernBERT text encoder, custom CrossChannelFormer fusing 5 fluorescence channels, and Cross-Well Alignment for batch-effect correction; enabled joint training on drug, CRISPR, and ORF perturbations
- Compressed 3.31 TiB of raw microscopy into 153 GB of cached DINOv3 embeddings, cutting per-epoch training to 3–5 min on single RTX 5080 across 3M+ images, 303 compounds, and 160 genes from CPJUMP1 benchmark

Hybrid ResNet-ViT for Skin Cancer Classification

Published in IEEE

July 2025

- Hybrid TensorFlow architecture combining frozen ResNet50 with four-head Vision Transformer blocks and Global Average Pooling for seven-class skin lesion classification, achieving 96.3% accuracy and macro F1 of 0.961 on HAM10000 (10,015 images, Albumentations preprocessing); presented at IEEE World Conference to 100+ attendees